

Lecture 2.2 – Basic Plotting with R

Markdown

Specific Learning Objectives:

2.2.1 – Create reproducible scripts in R.

2.2.2 – Include effective documentation in scripts and projects.

2.2.4 – Create and use Notebooks and documents using RMarkdown.

3.2 – Learn how to plot quickly using R's base graphics.

Skill Check 1 Debrief

- Stats presented in R Markdown Preview.
- **Problem areas:** (< 50% correct)
 - Q5 (Lists)

Question 5: Lists

I have created a list stored with the name `my_list` with the following code:

```
my_list <- list("flower" = c(1, 4.9, 18, 38, 20, 10.02),  
               "micha" = matrix(runif(30), nrow = 5),  
               "color" = data.frame("x" = c(1.1, 1.2, 1.3),  
                                     "y" = c(2.3, 5.6, 6.2)),  
               1:40)
```

How many members does this list contain? What data class is each member? How would you reference the integer 3 in `my_list`?

Check Your Understanding

Make a plot of vapor pressure in mmHg versus temperature in C of mercury using the pressure data set. Set the axis labels and title.

Make the same plot using a different method (either with or without a formula).

Check Your Understanding

Make a plot of circumference versus age of the orange trees in the data set Orange. Give each tree a unique color *and* point shape.

Check Your Understanding

Add a legend to your graph of Orange trees.

In-class Exercises

- 1. Pick a data set from the base package (`data()`) and make one or two plots. Add these plots to the end of your `L2.2Notebook.Rmd` file.**
- 2. Work on Assignment 2.1. (Remember, this should be a separate Rmd file than the lecture notes!)**

Action Items

- 1. Complete Assignment 2.1 using R Markdown.**
- 2. Read Davies Chapter 8 and Chang Chapters 1-2 for next time.**